



Estimating Magnitude Completeness in Earthquake Catalogs: A Comparative Study of Catalog-based Methods

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1. Motivation and methods

The purpose of the present paper is to assess the accuracy and reliability of commonly used **catalog-based methods** for estimating M_c using both simulated and observed data that reflect the heterogeneous temporal and spatial distribution of M_c . We introduce three new catalog-based methods for estimating M_c , drawing on our expertise in both statistical testing and computer science, and compare them with other existing methods. Additionally, we propose a novel seismicity pruning framework that not only simulates incomplete seismicity through a pruning function or the ANgular model, but also incorporates the modeling of the spatiotemporal distribution of this incompleteness. By examining the limitations of these foundational catalog-based methods in estimating M_c , we aim to refine our understanding of their appropriate applications, ultimately improving insights into seismicity patterns, earthquake forecasting, and hazard mitigation by emphasizing the importance of data-quality considerations.

- MAXC (Wyss et al., 1999): maximum curvature
- GFT-95% (Wiemer and Wyss, 2000): goodness-of-fit tests
- MBS-CG (Cao and Gao, 2002): method of b -value stability
- MBS-WW (Woessner and Wiemer, 2005): method of b -value stability
- EMR (Woessner and Wiemer, 2005): entire magnitude range
- MBASS (Amorèse, 2007): median-based analysis of the segment slope
- KST-95%: Kolmogorov-Smirnov tests
- BSReLU: Black-Scholes Rectified Linear Unit
- AEReLU: asymmetric exponential Rectified Linear Unit

2. AEReLU and BSReLU

$$\text{CCFMD}(m) = 10^{a-b \cdot G(m)}$$

in which,

$$G(m) = m_c + \sigma_c \cdot \text{gReLU}(x)$$

where $x = (m - m_c)/\sigma_c$

$$\text{BSReLU}(x) = \left(x + \frac{M_c}{\sigma_{M_c}} \right) \Phi \left(\frac{\ln[(x\sigma_{M_c} + M_c)/M_c] + \frac{\sigma_{M_c}}{2}}{\sigma_{M_c}} \right) - \frac{M_c}{\sigma_{M_c}} \Phi \left(\frac{\ln[(x\sigma_{M_c} + M_c)/M_c] - \frac{\sigma_{M_c}}{2}}{\sigma_{M_c}} \right)$$

$$\text{AEReLU}(x) = \begin{cases} \frac{e^{-\beta x}}{1+\beta} & x < 0 \\ x + \frac{e^{-\beta x}}{1+\beta} & x \geq 0 \end{cases}$$

- m_c : replaces the completeness magnitude as a transition magnitude around which earthquake detection change rapidly.
- σ_{M_c} : quantifies the magnitude range over which this transition occurs.
- β : controlling the degree of asymmetry

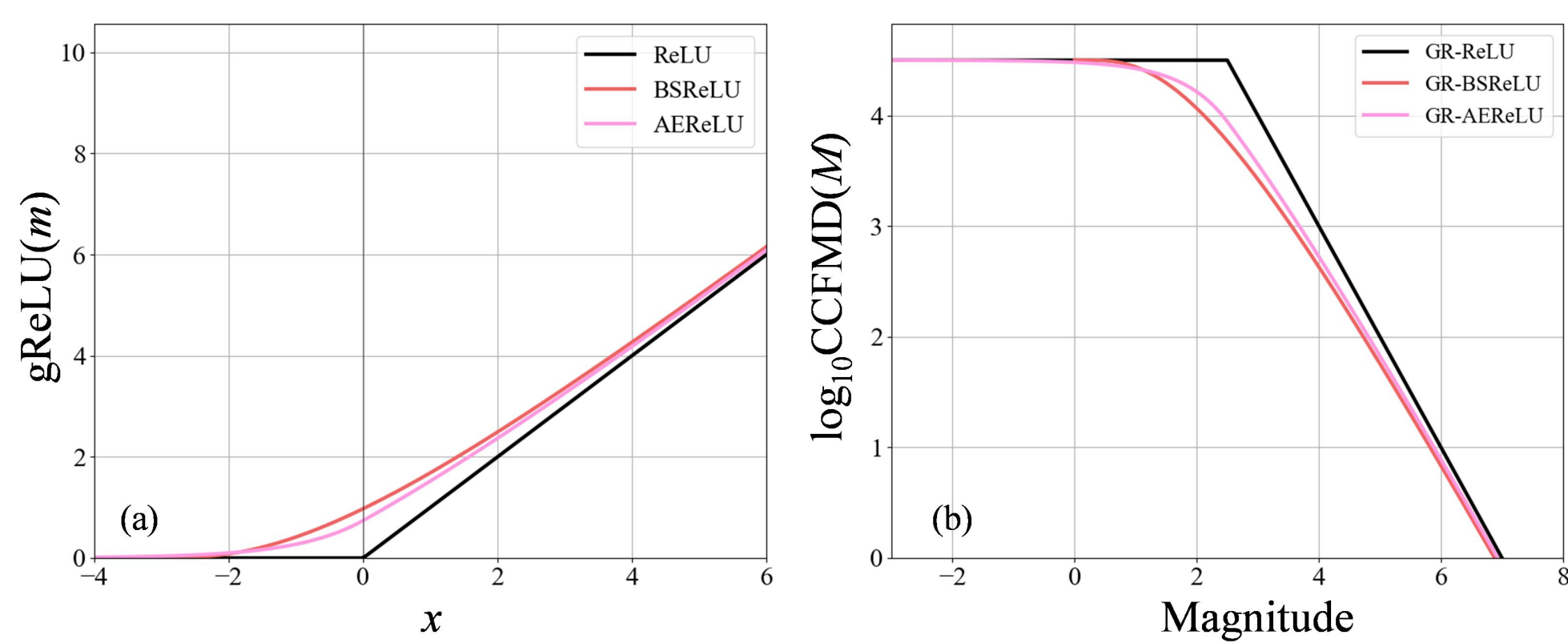


Figure 1. Plots of (a) the two $\text{gReLU}(x)$ functions introduced in the text and (b) the corresponding augmented complementary cumulative frequency-magnitude distributions (CCFMDs). For comparison, the figure also includes the Rectified Linear Unit, $\text{ReLU}(x)$, depicted with a black solid line, which is defined as $\text{ReLU}(x) = 0$ for $x < 0$ and $\text{ReLU}(x) = x$ for $x \geq 0$, along with the corresponding CCFMD recovering the pure Gutenberg-Richter law. The relationship between x in (a) and the magnitude m in (b) is given by $x = (m - M_c)/\sigma_{M_c}$. The parameters are set to $a = 7.0$, $b = 1.0$, $M_c = 2.5$, $\sigma_{M_c} = 0.75$, and $\beta = 0.3$, which are values that are informed by empirical data.

3. Earthquake catalog

Simulated earthquake catalog

- Modeling incompleteness with iterative pruning

$$y(m) = c_1 m + c_2, \quad m_{\min} \leq m < m_{\max}$$

- Modeling incompleteness with the AN (Angular) detection model

$$p(m) = \exp[-\kappa(m_{\max} - m)], \quad m_{\min} \leq m < m_{\max}$$

Observed Catalogs

- 6 Seismic Regions:

China: Beijing-Tianjin-Hebei, the southeastern coast, Sichuan-Yunnan, and northern Xinjiang

Global: California, New Zealand

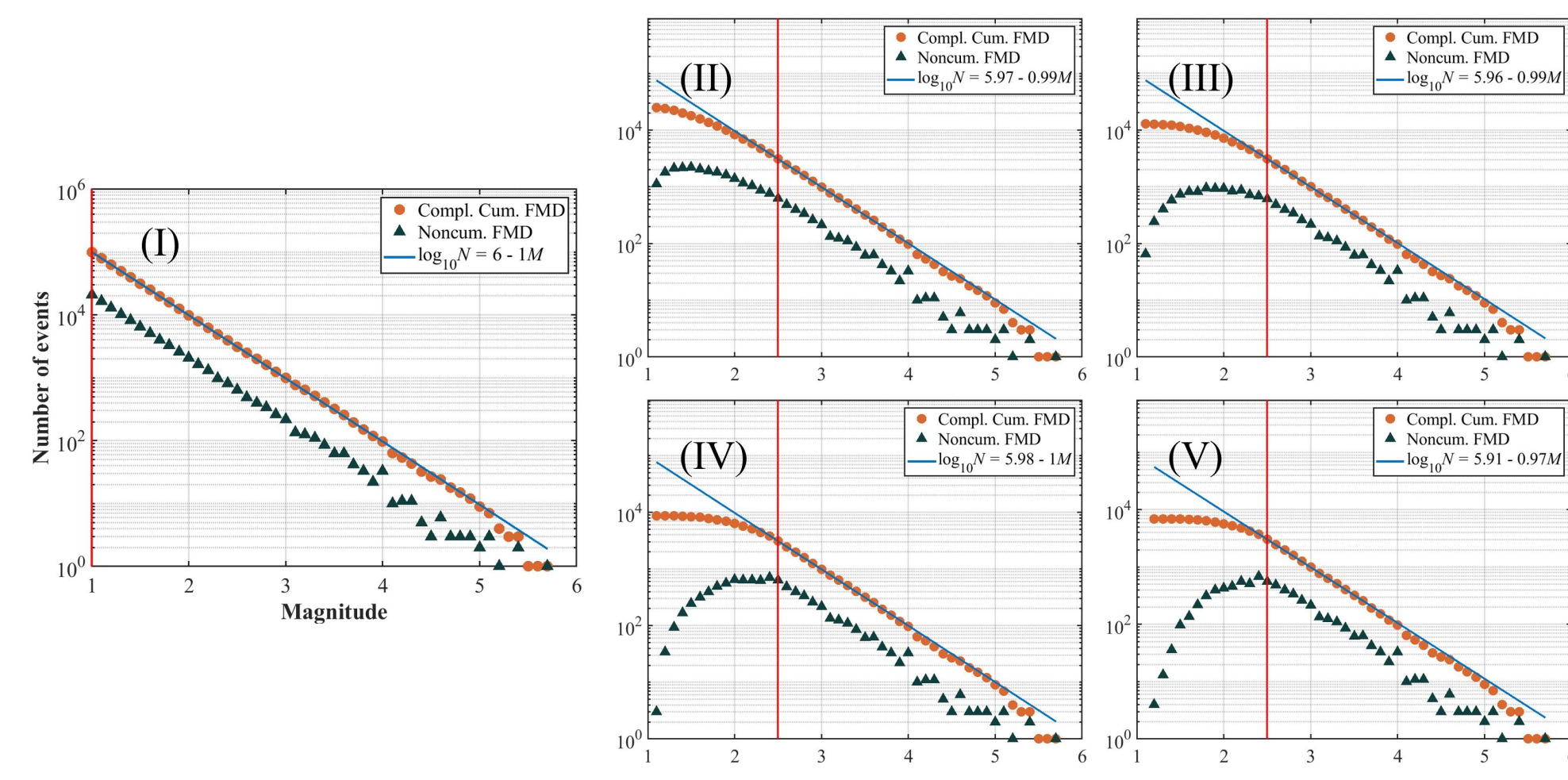


Figure 2. Complementary cumulative distribution (CCFMD) and density distribution (FMD) alongside the Gutenberg-Richter (GR) law, fitted using simulated earthquakes with magnitudes larger than 2.5. The total number of simulated earthquakes with $M \geq M_{\min} = 1.0$ is 10^5 . For each case, the b -value of the GR law is set to 1.0 to simulate synthetic seismicity, and panel (I) shows the CCFMD and FMD, which follow a pure power law for the entire magnitude range from $M_{\min}$ to $M_{\max}$. Panels (II) to (V) show results from iteratively applying the pruning function to magnitudes from $m_{\min} = 1.0$ to $m_{\max} = 2.5$, one to four times, respectively. By setting $M_{\min}^{\text{pruned}} = 2.5$, the data are separated into incomplete and power law (complete) parts. In this study, $m_{\max}$ (i.e., $M_{\max}^{\text{pruned}}$) is varied between 1.0 and 5.0, and each data set, is generated 200 times.

4. Estimating M_c for simulated seismicity with homogeneous M_c

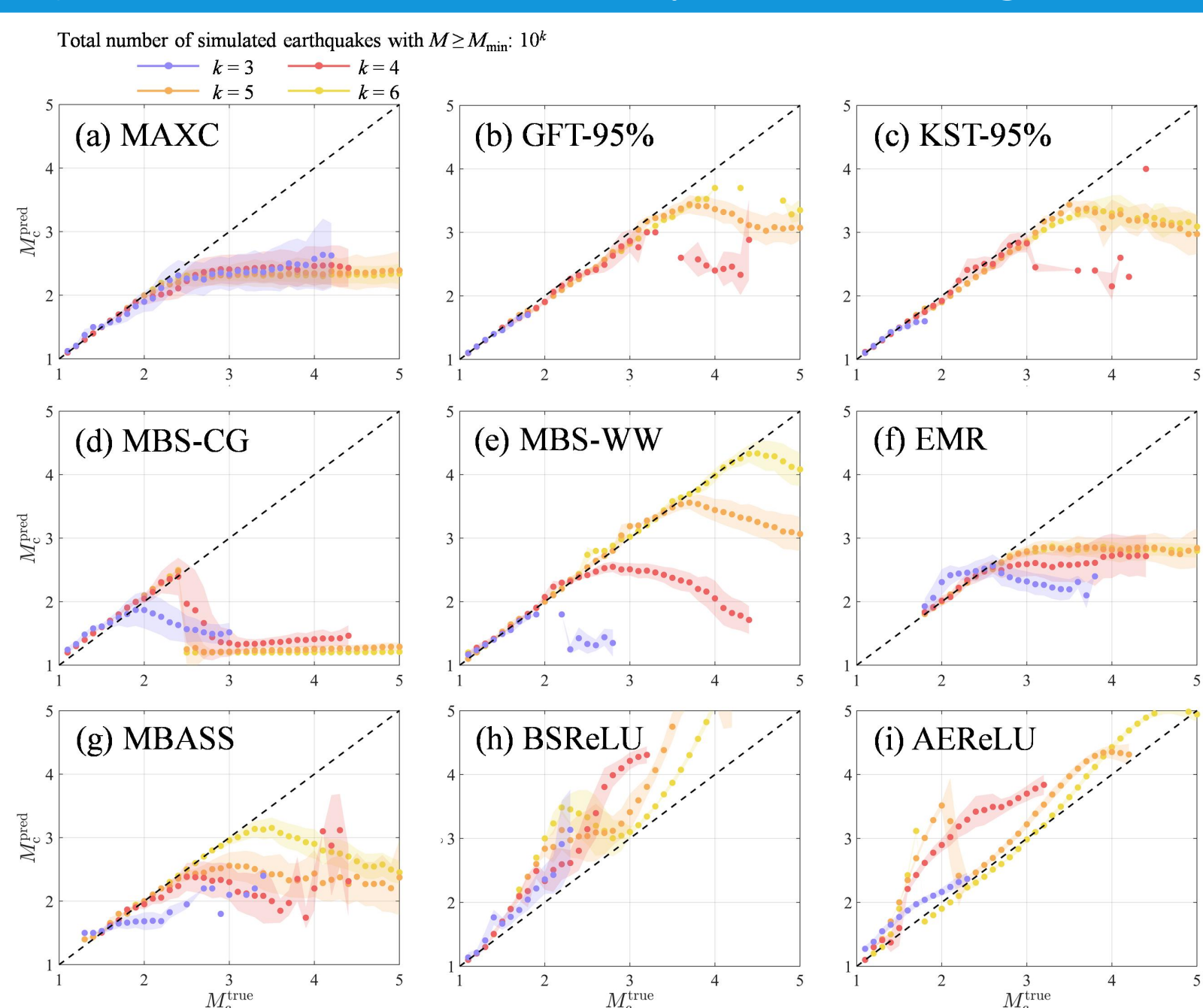


Figure 3. Results of the estimation of the completeness magnitude M_c^{pred} as a function of the true completeness magnitude M_c^{true} for the simulated datasets with homogeneous M_c after being iteratively pruned three times, as shown in Panels (IV) of Figure 2. The shaded areas represent the uncertainties in the estimated completeness magnitude M_c^{pred} . In panels (h) BSReLU and (i) AEReLU, M_c^{pred} is defined as the magnitude corresponding to a completeness probability of approximately 84%. The different colors correspond to the total number of simulated earthquakes with $M \geq M_{\min} = 1$, with 10^k for $k = 3, 4, 5, 6$.

5. Estimating M_c for simulated seismicity with heterogeneous M_c

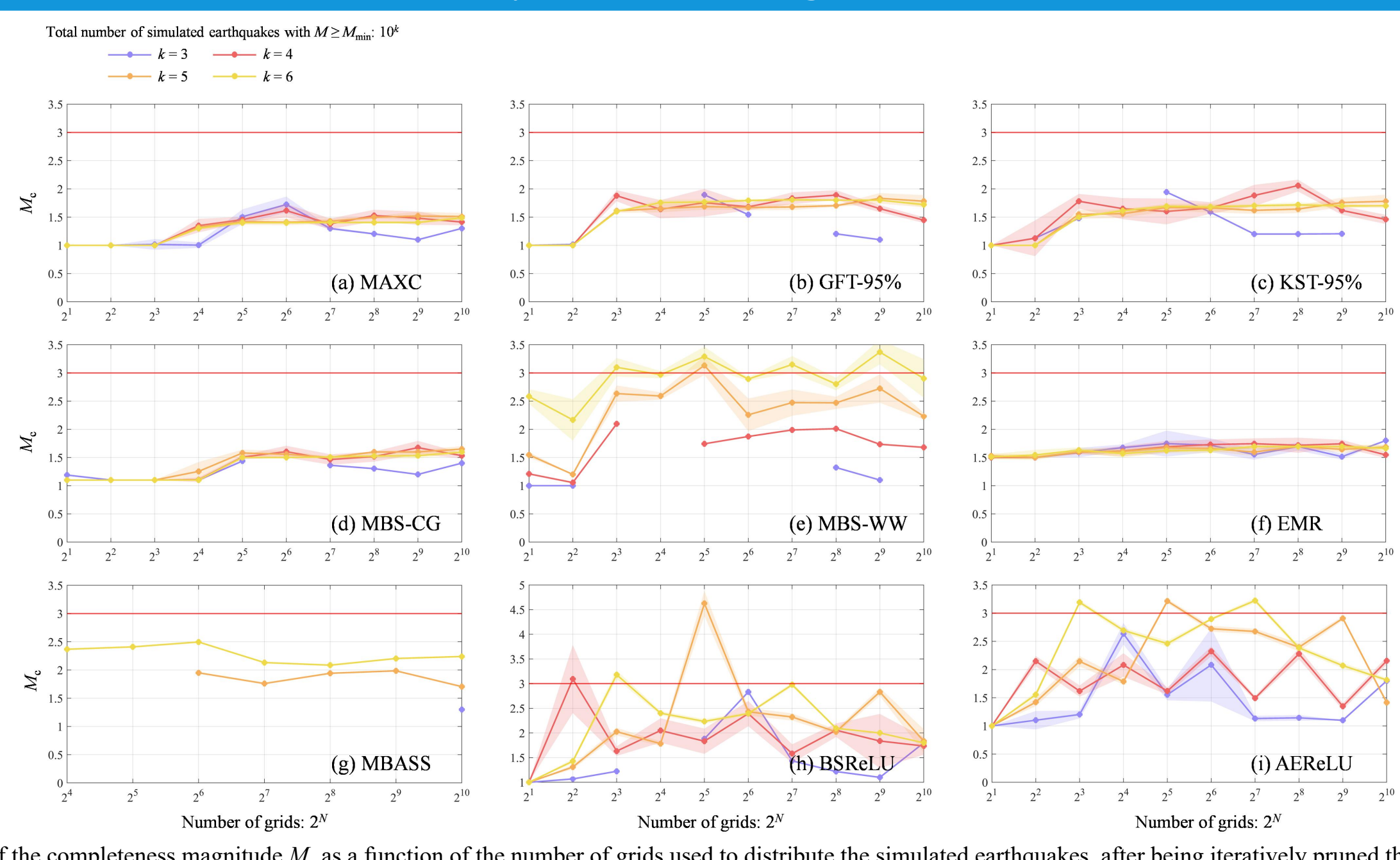


Figure 4. Results of the estimation of the completeness magnitude M_c as a function of the number of grids used to distribute the simulated earthquakes, after being iteratively pruned three times. The total number of simulated earthquakes with $M \geq M_{\min}$ given by 10^k for $k = 3, 4, 5, 6$, is randomly and uniformly distributed across 2^N ($N = 1, 2, \dots, 10$) grids. The minimum completeness magnitude ($M_c^{\min} = 1.0$) and maximum completeness magnitude ($M_c^{\max} = 5.0$) are assigned to the first and last grids, respectively, with the remaining grids having completeness magnitudes linearly interpolated between these values. For example, with $2^2 = 4$ grids, the completeness magnitudes are $M_c^{\text{grid}1} = 1.0$, $M_c^{\text{grid}2} = 2.3$, $M_c^{\text{grid}3} = 3.7$, and $M_c^{\text{grid}4} = 5.0$, ensuring that the average M_c across all grids is 3.0 (indicated by the red horizontal line in the figure). The shaded areas represent the uncertainties in the estimated completeness magnitude. The different colors correspond to the total number of simulated earthquakes.

6. Estimating M_c for empirical seismicity

Table 1. Estimation results of the completeness magnitude M_c and their associated uncertainties derived from 200 bootstrapping iterations, using seven methods based on the “cut-off M_c ” concept across the six study regions.

Region	MAXC	GFT-95%	KST-95%	MBS-CG	MBS-WW	EMR	MBASS
Beijing-Tianjin-Hebei	0.49 ± 0.10	1.01 ± 0.03	1.00 ± 0.01	0.86 ± 0.16	1.75 ± 0.17	0.80 ± 0.08	1.32 ± 0.64
Southeastern coastal	0.26 ± 0.06	0.92 ± 0.05	0.88 ± 0.05	0.34 ± 0.06	1.44 ± 0.20	0.60 ± 0.01	1.35 ± 0.63
Sichuan-Yunnan	1.10 ± 0.00	1.30 ± 0.01	1.30 ± 0.01	0.20 ± 0.00	1.71 ± 0.19	1.30 ± 0.00	1.60 ± 0.08
Northern Xinjiang	1.36 ± 0.06	1.59 ± 0.03	1.60 ± 0.05	0.20 ± 0.00	1.87 ± 0.32	1.57 ± 0.08	1.70 ± 0.07
California	0.90 ± 0.00	0.91 ± 0.00	0.91 ± 0.06	0.70 ± 0.00	1.84 ± 0.10	0.96 ± 0.10	0.91 ± 0.09
New Zealand	1.94 ± 0.05	2.17 ± 0.00	2.10 ± 0.00	0.20 ± 0.00	3.13 ± 0.35	2.00 ± 0.05	2.17 ± 0.05

Table 2. Estimation results of the parameters of BSReLU and AEReLU, along with their associated uncertainties derived from 200 bootstrapping iterations across the six study regions. Recall that in both BSReLU and AEReLU, M_c^{pred} is defined as the magnitude corresponding to a completeness probability of approximately 84%.

Region	BSReLU					AEReLU					M_c^{pred}
	a	b	M_c	σ_{M_c}	M_c^{pred}	a	b	M_c	σ_{M_c}	β	
Beijing-Tianjin-Hebei	5.85 ± 0.42	1.03 ± 0.07	0.62 ± 0.43	0.73 ± 0.21	1.97 ± 1.69	5.56 ± 0.12	0.97 ± 0.04	0.76 ± 0.09	0.38 ± 0.04	0.56 ± 0.11	1.65 ± 0.47
Southeastern coastal	5.62 ± 0.78	1.05 ± 0.11	0.74 ± 0.62	0.93 ± 0.28	2.03 ± 1.97	5.06 ± 0.13	0.96 ± 0.05	0.58 ± 0.10	0.32 ± 0.06	0.51 ± 0.10	1.35 ± 0.70
Sichuan-Yunnan	6.79 ± 0.02	0.92 ± 0.02	1.10 ± 0.01	0.95 ± 0.07	3.84 ± 0.53	6.72 ± 0.14	0.87 ± 0.01	1.09 ± 0.16	0.39 ± 0.02	0.39 ± 0.72	2.23 ± 0.22
Northern Xinjiang	6.08 ± 0.09	0.95 ± 0.03	1.36 ± 0.07	0.62 ± 0.09	2.83 ± 0.63	6.14 ± 0.14	0.92 ± 0.06	1.47 ± 0.08	0.39 ± 0.03	0.11 ± 0.20	4.08 ± 0.47
California	6.94 ± 0.38	1.23 ± 0.08	1.10 ± 0.31	0.88 ± 0.17	3.94 ± 2.23	6.55 ± 0.14	1.17 ± 0.04	0.96 ± 0.09	0.41 ± 0.05	0.46 ± 0.07	2.37 ± 0.47
New Zealand	7.45 ± 0.09	1.09 ± 0.02	1.95 ± 0.05	0.37 ± 0.04	2.91 ± 0.29	7.63 ± 0.10	1.11 ± 0.02	2.07 ± 0.06	0.39 ± 0.01	0.24 ± 0.16	4.06 ± 0.22

7. Key Points:

- Evaluated nine catalog-based methods for estimating the completeness magnitude (M_c) in both simulated and observed earthquake catalogs.
- Introduced BSReLU and AEReLU as augmented Gutenberg-Richter models, introducing a novel probabilistic framework for modeling the magnitude M_c of completeness.
- Demonstrated that AEReLU is the most reliable method even under the assumption of a sharp cut-off M_c , despite being designed for a gradual probabilistic transition from incomplete to complete detection.



paper: <https://arxiv.org/abs/2506.16849>

paper(GR-gReLU): doi.org/10.48550/arXiv.2506.16849

Code & Data: <https://github.com/lijiawei098/GR-gReLU>